

Review of FedImpute

I recommend that this paper should be published in Journal of Statistical Software, after some minor revisions. I give my remarks in the following.

Major comments

1. The authors place a lot of emphasis on missing data mechanisms, but only one imputation method supported by FedImpute actually handles the MNAR case (`notmiwae`). Of course, I don't expect the authors to add methods that handle the MNAR case, but this should be noted somewhere. The fact that FedImpute allows you to generate missing data of different types and with such diversity is already a contribution in my opinion.
2. Is it easy to change the hyperparameters of the methods? If so, I think the authors should highlight this flexibility more. If not, it should definitely be noted somewhere. There are methods (e.g. `miwae`) that are very sensitive to the choice of hyperparameters (e.g. the dimension of the latent variable).
3. FedImpute allows the user to have a real federated dataset that contains real missing data. From my understanding, in order to obtain imputation scores, the authors add synthetic missing data on which the scores are calculated. I think this should be explained in a little more detail, particularly the `create_real_scenario()` module, which could be better presented in the main text.

Minor comments

1. Page 12: the authors should specify which EM algorithm is used as an imputer (in Table 3, for example). To my knowledge, there are many EM algorithms for handling missing values depending on the statistical model.
2. Typo page 18: MANR
3. Page 22: what is `imp_ws` ? Wasserstein distance ?
4. As I understand it, FedImpute does not provide multiple imputation, although it uses several multiple imputation methods for imputers (such as `mice` and even `miwae`), this should be clarified somewhere.

5. Page 27: in Section 5.1, there are not many details about the missing data, in particular I do not see any mention of the percentages of missing data (only *under heterogeneous MNAR missing data scenario* page 28).
6. In section 5.2, if I understand correctly, additional missing values are added to calculate imputation measures (RMSE, etc.). What are the chosen missing value ratios?
7. It is a pity that only continuous variables are supported, whereas some imputation methods support both continuous and categorical variables. However, I understand the authors' choice. I hope Fedimpute will be upgraded later!
8. Can test sets contain missing data or not? This should be indicated.